Project Report: Microcontroller-Based Voltage Protector

1. Design Approach

The objective of this project was to design and simulate a microcontroller-based voltage protector that safeguards electrical appliances against undervoltage, overvoltage and overcurrent conditions. The system was developed in Proteus using an Arduino Uno as the controller. A voltage-sensing circuit monitors the supply voltage, while an ACS712-5A current sensor measures load current. A relay disconnects the load whenever a protection fault is detected.

2. Implementation Process

The circuit was designed and tested in Proteus. The voltage-sensing circuit was calibrated and using the rectified output the input voltage is measured and overvoltage/undervoltage fault is detected.

The ACS712 sensor was connected to another analog input for current monitoring. The relay was connected to digital pin D9, with HIGH switching the load ON and LOW switching it OFF using a relay.

The Arduino firmware implements voltage and current monitoring. After a fault occurs, the load remains disconnected until the measured conditions return to the permitted range and the user acknowledges the fault using the reset button or computer command.

A 16×2 LCD and a keypad provide local monitoring and threshold adjustment. Manual ON/OFF push buttons provide direct load control.

A Proteus Virtual Terminal enables computer communication, status monitoring and threshold configuration through serial commands.

3. Results and Demonstration

The completed simulation demonstrates automatic load disconnection under detected protection faults, manual load control, fault acknowledgement and adjustable protection thresholds. The LCD displays measured voltage, current, load state and configured limits. The keypad and display allow users to modify the settings without changing the firmware.

The virtual control works perfectly and threshold can be set from there also.

4. Conclusion

The project demonstrates a configurable, microcontroller-based appliance protection system with both local and computer interfaces.